



BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, Pilani
Pilani Campus
AUGS Division

First SEMESTER, 2026-2027
COURSE HANDOUT

Date: 08.08.2026

In addition to part I (General handout for all courses appended to the timetable), this portion gives further specific details regarding the course.

Course No.	: CHEM U101
Course Title	: <i>Atomic Structure, Bonding and Spectroscopy</i>
INSTRUCTOR-IN-CHARGE	: Pritam Kumar Jana
Instructor(s)	: Pritam Kumar Jana, Saumi Ray
Tutorial Instructor(s)	: Ram Kinkar Roy, Saumi Ray, Anil Kumar, Prashant U. Manohar, Madhushree Sarkar, Satyajit Patra, Avik Kumar Pati, Nitika Grover, Anish Rao

1.Course Description: Quantum theory and atomic structure, overview of bonding theories, bonding in metal complexes, diatomic molecules, polyatomic molecules, and in solids, rotational spectroscopy, vibrational spectroscopy, electronic spectroscopy, nuclear magnetic resonance spectroscopy, molecular structure determination using spectroscopic data.

2.Scope and Objective of the Course: Conceptual understanding of the quantum description of atom, development of an understanding of how atoms form molecules, introduction of how electromagnetic radiation interacts with matter and how this interaction forms the basis of spectroscopy, and how spectroscopy reveals molecular structure.

3. Text Books

T1: P. Atkins & J. De Paula, Elements of Physical Chemistry, 6th Edition, Oxford University Press.

4. Reference Books

R1: J. D. Lee, Concise Inorganic Chemistry, 5th Edition, Blackwell Science, Oxford, 1996.

R2: T. W. Graham Solomons, Craig B. Fryhle, and Snyder Scott A., Organic Chemistry, Global Edition

5. Course Plan

Module	Lectures	Lect. No.	Reference	Learning Outcome
Module 0: Introduction of the course (1 lecture)	Atom to molecular structure determination	1	-	Course overview and requirement of subject
Module 1: Quantum Foundations of Atomic Structure (8 lectures)	Review of failure of classical theory, Wave-particle duality, uncertainty principle, Schrödinger equation	2-3	T1: 12.1-12.6	Recognize the need for quantum mechanics, Consolidate new concept to use in quantum mechanics
	Some exactly solvable problems: Particle-in-a-box model, Particle on a ring, Spherical harmonics	4-5	T1: 12.7-12.8	Clarification of quantization of energy, Zero-point energy, Description of conjugated systems, Quantum dots
	Hydrogen atom, Many electron atoms, Term symbols	6-9	T1: 13.1-13.7	Atomic orbitals, Quantum number, orbital shapes, Term symbols in



BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, Pilani
Pilani Campus
AUGS Division

				relation to atomic spectra for example Na D-line
Module 2: Chemical Bonding in Molecules (9 Lectures)	Valence bond theory (review only/self-study), Molecular orbital (MO) theory	10-11	T1: 14.3-14.11	MO diagrams for diatomic molecules (Homonuclear and heteronuclear (max two examples such as CO and HF))
	Hückel and frontier molecular orbital approaches, Band theory	12-14	T1: 14.12-14.13 T1: 17.1-17.2	Bonding in polyatomic molecules, Bonding in solids, Delocalized bonding, Frontier molecular orbital description of acyclic and cyclic π -conjugated molecules, Connection to chemical reactivity (Example- Diels-Alder reaction), Extension of molecular orbitals to solids, Band formation, Conductors, Semiconductors, Superconductor, Insulators
	Crystal field theory, Bonding in metal complexes	15-18	R1: pg. 203 – 22	Coordinate bonding, Crystal field splitting of d orbitals, Ligand field effects and distortions, Applications relevant to MOFs and bio-metal complexes
Module 3: Understanding of Molecular Structure through Spectroscopy (10 Lectures)	Rotational, Vibrational, Raman, Electronic spectroscopy	19-23	T1: 12.9, 19.6-19.11, 19.13, 20.1-20.4	Rotational spectroscopy of diatomic molecules, Molecular vibrations, Harmonic oscillator model and anharmonicity (qualitative only), Selection rules and determination of bond strength of molecule, IR spectra of simple molecules, Basic concepts, Selection rules and mutual exclusion principles for Vibrational Raman spectroscopy, Electronic transitions (absorption and emission), Lambert-Beer's Law in the context of electronic spectroscopy, Selection rules, Electronic spectra of conjugated molecules and metal complexes
	Nuclear Magnetic Resonance (NMR) Spectroscopy	24-25	R2: 2.16, R1: 21.1-21.4 R2: 9.1-9.8	Nuclear spin and magnetic moments, Interaction with external magnetic field, Resonance condition, Chemical shift, Spin-spin coupling, Basic idea of structure determination with proton NMR



BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, Pilani
Pilani Campus
AUGS Division

	Case Studies: Molecular structure determination by combining vibrational (IR), proton NMR and electronic spectra	26-28	-	Using spectroscopic method identification of the structure of molecules
--	--	-------	---	---

6. Evaluation Scheme:

Component		Duration	Weightage (%) (Total Marks 200)	Date & Time	Remarks
Mid-Semester Test		90 mins	25 (50 marks)	05/10 AN2	Closed Book
Continuous Evaluation	Standard evaluations#	-	25 (50 marks)	Continuous	Closed Book
	Attendance linked marking scheme*	-	10 (20 marks)	Continuous	Closed Book
Comprehensive Exam		180 mins	40 (80 marks)	03/12 AN	Partially open book

Standard evaluations

- A total of **three quizzes (each of 25 marks)** will be conducted under continuous evaluation. **The best two** will be considered for final evaluation (i.e., one buffer).
- Only textbooks, reference books, class/tutorial hand-written notes, and course material (if any provided) will be allowed in the open book examination; only scientific non-programmable calculators are allowed during the tutorial, mid-semester, and comprehensive examinations.

* Attendance linked marking scheme

- Three tests (each of 8 marks)** will be conducted during the tutorial hours or along with the standard evaluations, and the **best two scores** will be considered for evaluation. A student's test marks will be counted only if they maintain **at least 60% attendance in the tutorial classes**.
- The remaining **2% (4 marks)** of the total course marks will be awarded to students who maintain **at least 60% attendance in the lecture classes**.

7. Chamber consultancy hours: To be announced separately.

8. Notices: Notices, if any, will be uploaded/ displayed in the **Nalanda only**.

9. Make-up Policy: (Please also refer to Part I of the handout given in the timetable booklet.) Make-up for quizzes is highly discouraged, as sufficient buffers are provided to the students. Make-up for the mid-semester test and the comprehensive exam may be considered for genuine cases as per the Institute's rules **against an application** by the student with documents.

10. Minimum performance requirement for valid grade: To clear the course, a student should obtain 30% of the average marks of the top 10 performers in the class, or 40% of the median (whichever is lower). **If any student gets the marks lower than the prescribed standard mentioned above, the student may be awarded NC.**

11. Attendance Policy: Students are expected to attend all classes to understand the course better.

Instructor-in-charge
CHEM U101